

Delta Hedging under Stochastic Volatility

When Structure Beats Learning

Stochastic Calculus and Its Applications in Finance

Aditya Nagarsekar, M P Ashish Bhat

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The Delta Hedging Problem

Goal: Hold a short call; dynamically trade the underlying to offset delta risk.

BSM textbook solution: Under frictionless continuous trading,

$$\Delta^* = \Phi(d_1), \quad d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}$$

replicates the option perfectly.

But practice violates two key assumptions:

- 1 Trading is *discrete* and *costly* — proportional TC $\kappa|\Delta\delta| \cdot S$
- 2 Volatility is *unknown* and *stochastic*

Research Questions

- Can RL learn a better hedge than BSM?
- Does an LSTM vol forecast improve hedging?
- How do they combine?

Setting

Nifty 50 options, 2010–2024

51 non-overlapping 21-day test windows

Transaction cost $\kappa = 0.001$

Mathematical Framework: Risk-Neutral Pricing

Under the physical measure $\mathbb{P}$, the stock follows GBM:

$$dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbb{P}}$$

Girsanov's theorem (change of measure):

$$\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{\mathcal{F}_T} = \exp\left(-\theta W_T^{\mathbb{P}} - \frac{1}{2}\theta^2 T\right), \quad \theta = \frac{\mu - r}{\sigma}$$

Under $\mathbb{Q}$, the discounted stock $e^{-rt}S_t$ is a *martingale*, and the fair option price is:

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+]$$

BSM PDE (Itô's Lemma + No-Arbitrage)

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

CRR Binomial Tree

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = 1/u$$

$$p = \frac{e^{r\Delta t} - d}{u - d}, \quad N = 100$$

Converges to BSM as $N \rightarrow \infty$

Hedging Error Decomposition

Under discrete rebalancing, the one-step hedging error can be decomposed:

$$\varepsilon_t = \underbrace{(\delta_t - \Delta_t^*) \Delta S_t}_{\text{controllable: hedge gap}} - \underbrace{\frac{1}{2} \Gamma [(\Delta S_t)^2 - \sigma^2 S_t^2 \Delta t]}_{\text{irreducible: gamma noise}}$$

Controllable term

How far is implemented hedge δ_t from analytic target Δ_t^* ?

Lever 1: Better σ estimate $\Rightarrow$ better Δ^*

Lever 2: Better execution rule

Gamma noise variance

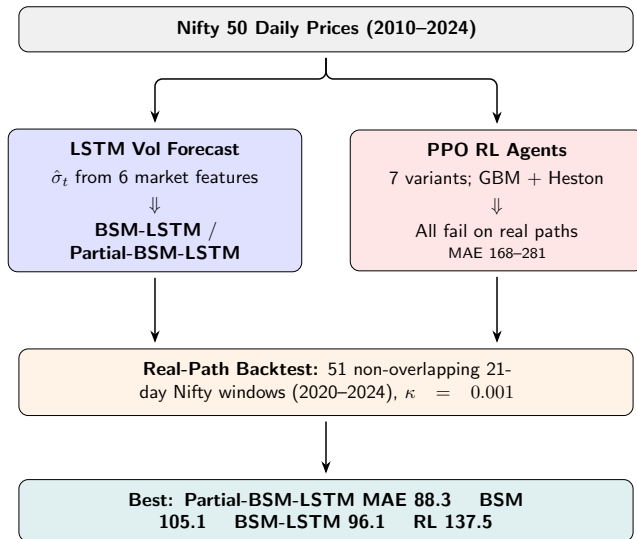
$$\text{Var}(\text{gamma noise}) = \frac{1}{2} \sigma^4 S^4 \Gamma^2 (\Delta t)^2$$

Peaks ATM near expiry; expectation = 0 by Itô's isometry.

Irreducible with daily rebalancing.

Implication: RL must beat BSM on the *controllable* term — a hard task when BSM's $\Phi(d_1)$ is already analytically optimal under GBM.

Project Architecture



PPO Agent: Setup

Environment: Simulated GBM (and Heston) paths for a 21-day European call

State (base): $s_t = (\text{moneyness}, \tau/T, \delta_{t-1})$;
extended variants additionally include $\hat{\sigma}_t$ and BSM- Δ depending on the variant

Action: $a_t = \delta_t \in [0, 1]$

Reward (base):

$$r_t = -\varepsilon_t^2 - \kappa|\delta_t - \delta_{t-1}|S_t$$

Penalises squared one-step hedging error plus proportional TC

Itô-inspired variants subtract the gamma noise term; other variants add Γ as a state feature — explored across the 6 extensions

PPO Clipping

$$L^{\text{CLIP}} = \mathbb{E}[\min(r_t A_t, \text{clip}(r_t, 1 \pm \epsilon) A_t)]$$

Conservative updates prevent destructive policy changes in noisy financial rewards

6 Variants Tested

- Domain randomisation ($\sigma \sim U[0.10, 0.40]$)
- LSTM-conditioned PPO
- Anchor-controller (adjust around BSM Δ)
- Discrete DQN
- Heston environment
- Reward design variants

RL Results in Simulation

Agent (variant)	MAE	BSM- Δ Corr.
BSM (analytic)	0.284	—
A: GBM + MSE (base)	0.830	0.947
B: GBM + $\text{It}\hat{o}$ reward	0.926	0.935
C: Heston + MSE	0.790	0.963
D: Heston + $\text{It}\hat{o}$	0.790	0.958
E: GBM + $\text{It}\hat{o}$ + Γ obs	0.764	0.957
F: GBM + domain rand. σ	0.723	0.963

Natural habitat (varying- σ env, fair comparison):

- LSTM-RL MAE: **0.407** vs 1.435 constant- σ baseline
- -72% **MAE**, -64% **CVaR**₉₅ ($p < 0.001$, paired bootstrap)
- CVaR: -1.85 vs -5.08 — catastrophic tails eliminated

LSTM-guided BSM still wins

Even in the natural habitat, BSM-LSTM (MAE 0.227) outperforms LSTM-RL (0.407). The analytic formula re-evaluates $\Phi(d_1)$ at every step exactly; RL generalises from training imperfectly.

Utility dominance

$$(J_\lambda = \text{MAE} + \lambda \cdot \text{TC})$$

LSTM-guided BSM wins at *every* risk-aversion λ :

J_1 : **0.339** vs BSM 0.430 vs RL 0.895

J_5 : **0.786** vs BSM 1.015 vs RL 1.424

Volatility Forecasting: Why It Matters

The Vega link: A 1%-point error in σ propagates to a pricing error of

$$\text{price error} \approx \text{Vega} \times 0.01 = S\sqrt{\tau} \phi(d_1) \times 0.01$$

For ATM Nifty options ($S \approx 22,000$, $\tau = 21/252$): **≈ 25 points per 1 pp vol error.**

GARCH error ≈ 1.9 pp $\Rightarrow$ **49 pts**; LSTM ≈ 1.1 pp $\Rightarrow$ **29 pts**; rv21 ≈ 0.4 pp $\Rightarrow$ **11 pts**. GARCH $\rightarrow$ LSTM saves **≈ 20 pts/option**.

LSTM Architecture

- 2-layer stacked LSTM, hidden dims **64 + 32**
- Sequence length: **60 trading days**
- 6 features: rv5, rv21, rv_ratio, log-return, —return—, mom5
- Target: σ_{realized} over next 21 days
- Train 70% / Val 15% / Test 15% (chronological)

Key Feature: rv_ratio

$$\text{rv_ratio} = \frac{\text{rv5}}{\text{rv21}}$$

Encodes whether short-horizon vol is elevated vs. medium-horizon — a regime-change signal
GARCH has no equivalent for.

Volatility Forecasting Results

Model	RMSE	MAE	Dir. Acc.
LSTM	0.0203	0.0114	0.503
GARCH(1,1)	0.0254	0.0194	0.479
rv21 naive	0.0102	0.0043	0.544
India VIX	0.0453	0.0344	0.523

LSTM vs GARCH

41% lower MAE (0.0114 vs 0.0194)

20% lower RMSE (0.0203 vs 0.0254)

The rv21 paradox

A *naive persistence baseline* beats all models at the 21-day horizon.

Why? The trailing window $[t-20, t]$ overlaps **86%** with the forward target $[t+1, t+21]$ under strong volatility clustering (Nifty Hurst exponent ≈ 0.65), giving rv21 an inherent informational advantage. Any more complex model must improve on something already near-optimal. rv21 is a *transparency control*, not a model to beat.

Directional accuracy $\approx 50\%$

Persistence (predicts same direction) and mean-reversion (predicts opposite) cancel. Level prediction is the right metric.

India VIX and the Variance Risk Premium

Why does India VIX underperform?

India VIX is implied volatility — it is computed under the risk-neutral measure $\mathbb{Q}$:

$$\sigma_{\text{VIX}}^2 = \mathbb{E}^{\mathbb{Q}} \left[\int_0^T \sigma_t^2 dt \right] + \text{VRP}$$

The **Volatility Risk Premium** (VRP $\approx 4\text{--}6\%$ annualised) represents what investors pay for variance protection — a systematic upward bias of 4–6% over realised vol.

LSTM forecasts under $\mathbb{P}$ (physical measure); using VIX directly as a forecast overstates realised vol.

Girsanov in action

The market price of vol risk θ_v shifts $\mathbb{Q}$ away from $\mathbb{P}$:

$$\sigma_{\text{impl}} \approx \sigma_{\text{realised}} + \text{VRP}$$

Using implied vol without VRP correction is a *measure mismatch*.

Practical implication

India VIX would need a data-driven VRP correction before it can be used as a vol forecast for physical-measure hedging.

Option Pricing Horse-Race

Overall MAE (ATM synthetic, $T = 21/252$):

Method	MAE	Note
GARCH(1,1)	43.6	Over-reacts to spikes
LSTM	30.3	rv21 dominates at 21d
Historical (rv21)	18.6	Persistence effect
Implied (lagged)	18.7	Freshest market info

By moneyness (MAE, implied best everywhere):

	OTM	ATM	ITM
Historical	9.56	17.67	7.70
Implied	8.56	17.31	6.80
LSTM	14.27	25.88	11.37

Why rv21 $\approx$ implied in pricing

Both are slow-moving, high-overlap proxies for the next 21-day vol. The 86% window overlap makes trailing rv21 nearly as informative as yesterday's IV.

Why LSTM lags in pricing

At the 21-day horizon, rv21 persistence dominates. LSTM's advantage is in longer-horizon *regime* shifts; at this lag it adds less than persistence alone.

LSTM still beats GARCH by **31%** (30.3 vs 43.6).

Why LSTM still matters for hedging

Pricing uses fixed-point vol; hedging uses it *pathwise* — LSTM corrects local σ deviations step-by-step via the Vega link.

Real-Path Hedging Backtest

Setup: 51 non-overlapping 21-day windows on Nifty 50 (2020–2024), $\kappa = 0.001$

Strategy	Mean P&L	Std P&L	MAE [95% CI]	Mean TC
BSM daily (rv21)	+1.19	162.2	105.1 [74.7, 141.0]	36.0
RL PPO (rv21)	−13.4	205.0	137.5 [99.3, 181.1]	39.4
BSM-LSTM	+7.48	144.0	96.1 [69.6, 127.2]	35.6
Partial-BSM-LSTM	+1.02	134.6	88.3 [63.6, 118.4]	27.4

Best model wins

Partial-BSM-LSTM vs BSM:

- MAE: 105.1 → **88.3** (−16%)
- TC: 36.0 → **27.4** (−24%)

Tail risk: BSM $\text{CVaR}_{95} = -457$
vs RL = -535

RL has **17% heavier tails** on
real paths

Statistical caveat

BSM-LSTM vs Partial-BSM-LSTM: $p = 0.121$ (not significant at 51 windows). BSM vs RL: $p = 0.013$ — significant.

High-vol regime ($\sigma > 25\%$)

Regime	BSM	RL
Low (<15%)	95.6	117.9
Med (15–25%)	88.2	154.8
High (>25%)	384.5	377.7

high-vol domain randomisation helps

Why RL Fails on Real Paths

All pure-RL variants fail to outperform BSM:

RL Variant	Real-path MAE
Base PPO (rv21)	137.5
LSTM-conditioned PPO	168.2
Anchor-residual (LSTM)	214.9
DQN discrete (LSTM)	251.8
Anchor-fraction (DR/LSTM)	281.1
<i>BSM baseline</i>	<i>105.1</i>
<i>Hybrid BSM+DR-RL</i>	<i>100.0</i>

Hybrid shows promise but still beaten by Partial-BSM-LSTM (88.3)

Failure mode: underhedging

Policies converge to **low-turnover**

underhedging: reduce trading costs but allow large residual hedge error.

TC lower (39.4 vs 36.0) — policy learned to *trade less*, not hedge better.

Tail risk: RL $\text{CVaR}_{95} = -535$ vs BSM -457 — RL has **17% heavier real-path tails**.

Root cause: distribution shift

GBM training lacks real Nifty's volatility clustering, heavy tails, and regime changes.

Exception: in the high-vol regime ($\sigma > 25\%$), domain-randomised RL (MAE 377.7) marginally beats BSM (384.5) — the only real-path RL win.

The Itô Reward Paradox

Base (A): squared hedging error

$$r_t^A = -\varepsilon_t^2 - \kappa|\delta_t - \delta_{t-1}|S_t$$

Variant B: subtract gamma noise from reward

$$r_t^B = -\left(\varepsilon_t - \frac{1}{2}\Gamma[(\Delta S_t)^2 - \sigma^2 S_t^2 \Delta t]\right)^2 - \kappa|\delta_t - \delta_{t-1}|S_t$$

Intended to isolate the controllable error

$\varepsilon_t + G_t = (\delta - \Delta^*)\Delta S$; subtraction doubles the noise instead.

In practice, B is worse than A (MAE 0.926 vs 0.830).

Problem: Near convergence, gradient

$\propto (\varepsilon_t - G_t) \approx -2G_t$ — *pure noise*, destabilising training rather than converging.

Variant E: adds Γ as a *state observation* instead.

Gradient preserved; achieves MAE 0.764 — best among all variants without domain randomisation.

Why structure helps RL

Providing the gamma observation gives the agent structural guidance about curvature risk — information that pure reward shaping obscures.

Even with this, RL still fails to beat BSM on real paths.

Lesson

RL in finance is fragile to reward design. The Itô decomposition is mathematically elegant but doesn't straightforwardly translate to a better reward function.

Partial-BSM-LSTM: Design and Intuition

Executed hedge at time t :

$$\delta_t = \begin{cases} \delta_{t-1}, & |\Delta_t^* - \delta_{t-1}| \leq \theta \\ \delta_{t-1} + \alpha(\Delta_t^* - \delta_{t-1}), & \text{otherwise} \end{cases}$$

where $\Delta_t^* = \Phi(d_1)$ computed at LSTM forecast $\hat{\sigma}_t$.

Best hyperparameters: $\alpha = 0.9$, $\theta = 0.1$

Intuition from the decomposition:

- Small adjustments ($|\Delta^* - \delta| \leq \theta$) are dominated by TC + gamma noise $\Rightarrow$ hold
- Large deviations require correction $\Rightarrow$ partial step
- $\alpha < 1$ smooths the hedge path, further reducing TC

Why this beats full rebalancing

Full rebalancing (BSM-LSTM, $\alpha = 1$, $\theta = 0$) achieves MAE 96.1 but TC 35.6.

Partial rule: MAE **88.3**, TC **27.4**.

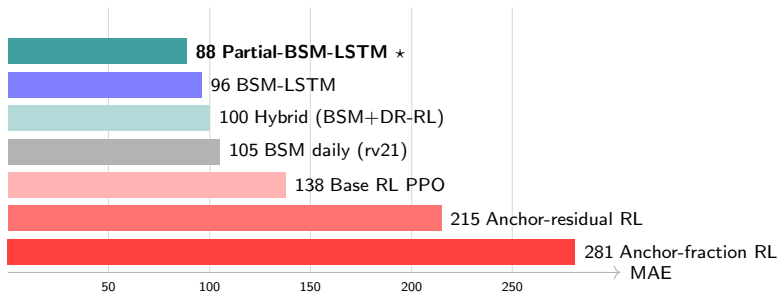
Reduced TC leaves more capital to absorb genuine hedging error.

vs. RL anchor-controller

RL also tried partial execution around a BSM anchor — but converged to a degenerate low-trading policy.

The key difference: *deterministic* partial rebalancing is transparent and doesn't overfit to simulation.

Performance at a Glance



LSTM + execution gains

LSTM vol input: 105 $\rightarrow$ 96 (-8.6%)

Partial execution: 96 $\rightarrow$ 88 (-8.1%)

Combined gain: 105 $\rightarrow$ 88 (-16%)

RL outcome

Most RL variants sit well above BSM; the hybrid (100) shows some promise but is still beaten by the deterministic partial rule (88).

Four Key Lessons

❶ **Volatility estimation dominates performance.**

BSM is already strong; improving σ from rv21 to LSTM produces a more reliable gain than replacing the hedge rule. Vega amplifies every vol forecasting improvement.

❷ **Execution strategy matters more than learning delta directly.**

Partial-BSM-LSTM outperforms full BSM-LSTM *not* by changing the hedge target, but by changing how aggressively the portfolio is moved toward it.

❸ **RL struggles because of noise and distribution shift.**

Policies trained under GBM do not transfer to real Nifty dynamics (clustering, heavy tails, regimes). More expressive simulation (Heston) helps slightly in-simulation but anchor-based RL variants fail even more severely on real paths.

❹ **Simple deterministic control around a strong analytic baseline beats RL.**

The best model in the study combines analytic structure, learned vol, and a transparent partial-adjustment rule. No end-to-end learning required.

Conclusion

Bottom line:

Structure + Good Inputs + Simple Control



Complex End-to-End Learning

BSM remains a very strong baseline. It can be improved:

- **Better inputs:** LSTM -41% MAE over GARCH; wins J_λ at every λ
- **Better execution:** Partial rule -24% TC, -16% MAE
- **Distributional alignment:** LSTM-RL -72% MAE and -64% CVaR₉₅ in natural habitat ($p < 0.001$)
- **Tail safety:** BSM CVaR -457 vs RL -535 on real paths

Future Directions

- **Richer simulation:** jump-diffusion, regime-switching GBM
- **VRP correction:** adjust LSTM forecasts toward $\mathbb{Q}$ measure
- **Longer test windows:** $n \gg 51$ for statistical significance of partial rebalancing gain
- **Model-based RL:** learn vol model directly from data, then plan

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Thank You

Delta Hedging under Stochastic Volatility:
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Partial-BSM-LSTM: MAE 88.3 TC 27.4 Best result

Aditya Nagarsekar